

Quadrilaterals — Full Test

Mixed questions from all five concepts. Section A is interactive; Sections B and C are written. Time: 45 minutes ·
Total: 28 marks. Answers on the separate key.

Objective (1 mark each)

Q1 · MCQ**[1]**

The sum of the interior angles of a quadrilateral is:

- (a) 180° (b) 360° (c) 540° (d) 720°

Q2 · MCQ**[1]**

A quadrilateral whose diagonals bisect each other at right angles is a:

- (a) rectangle (b) rhombus (c) trapezium (d) parallelogram

Q3 · MCQ**[1]**

The number of diagonals in a polygon of n sides is:

- (a) n (b) $n - 3$ (c) $n(n - 3)/2$ (d) $n(n + 1)/2$

Q4 · MCQ**[1]**

In a parallelogram, opposite angles are:

- (a) supplementary (b) equal (c) complementary (d) unequal

Q5 · FILL IN THE BLANK**[1]**

In quadrilateral ABCD, $\angle A = 100^\circ$, $\angle B = 80^\circ$, $\angle C = 70^\circ$. Then $\angle D =$ _____ degrees.

Q6 · TRUE / FALSE**[1]**

Every square is a rhombus.

True False

Q7 · MCQ

[1]

Which quadrilateral has exactly one pair of parallel sides?

(a) Trapezium (b) Parallelogram (c) Rhombus (d) Square

Q8 · ASSERTION-REASON

[1]

Assertion (A): The sum of the interior angles of a quadrilateral is 360° .

Reason (R): A quadrilateral can be divided into two triangles.

(a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q9

[2]

The angles of a quadrilateral are in the ratio 1 : 2 : 3 : 4. Find each angle.

Q10

[2]

The diagonals of a rhombus are 6 cm and 8 cm. Find the length of each side.

Q11

[2]

In parallelogram ABCD, $\angle A : \angle B = 2 : 3$. Find $\angle A$ and $\angle B$.

Q12

[2]

Find the length of the diagonal of a square of side 7 cm. (Use $\sqrt{2} \approx 1.41$.)

Long answer (3 marks each)

Q13

[3]

In trapezium ABCD, $AB \parallel DC$. If $\angle A = 75^\circ$ and $\angle B = 110^\circ$, find $\angle C$ and $\angle D$ with reasons.

Q14

[3]

One angle of a parallelogram is three times its adjacent angle. Find all four angles.

Q15

[3]

A regular polygon has each exterior angle equal to 40° . Find (a) the number of sides and (b) each interior angle.

Q16

[3]

The diagonals of a rectangle are $(2x + 1)$ cm and $(3x - 1)$ cm. Find x and the length of each diagonal.

Total: 28 marks · Check your work against the **answer key**.